

Question number	Answer	Additional guidance	Mark
7(a)(i)	16.82 / 16.83 / 16.87 / 16.8 / 16.9 / 17 (1)		(1)

Question number	Answer	Additional guidance	Mark
7(a)(ii)	A description that includes two of the following points: - used to make {amino acids / proteins} needed for growth (1) - used to make enzymes for chemical reactions - used to make DNA {which contains the genetic code / when (plant) cells divide} (1) - used to make ATP needed {as a source of energy / named process} (1) - used to make {chlorophyll / named light absorbing pigment} to <u>absorb</u> light for {light-dependent stage / photolysis} <u>(1)</u>	ACCEPT used to make amino acids to make {proteins / enzymes} used to make membrane transport proteins ACCEPT named enzyme in named chemical reaction ACCEPT used to make {bases / named base} to make {DNA / RNA / nucleic acids} ACCEPT used to make adenine to make {ADP / ATP}	(2)

Question number	Answer	Additional guidance	Mark
7 (b)(i)	74 (%) (1)	DO NOT ACCEPT any other valuw	(1)

Question number	Answer	Additional guidance	Mark
7 (b)(ii)	$1.5 \times 10^{12} / 1.54 \times 10^{12}$ (kg) (1)		(1)

Question number	Answer	Mark
*7(b)(iii)	Indicative content: Aspect 1 : Phytoplankton • more phytoplankton results in more photosynthesis • absorbing more carbon dioxide from the air • therefore less greenhouse gases • so less infra red radiation trapped • reducing global warming • release of nutrients into oceans will increase number of phytoplankton • because they use the nitrate for growth and division Aspect 2 : Animals other than whales • so there will be more carbon dioxide released into the air • due to respiration • which would increase global warming • increase in phytoplankton will increase numbers of whales / other animals Aspect 3 : Whales • more whales more carbon dioxide from respiration • more whales more nitrates released • there would be a lot of carbon stored in the whale's body • as {carbohydrate / protein / fat} • whale acts as a carbon sink • dead {whales / organisms} decompose very slowly • as conditions unfavourable for {decomposers / decomposition} • therefore this carbon will not be released back into the atmosphere as carbon dioxide • helping to reduce global warming Aspect 4 : Decomposition of phytoplankton and other animals • increase in phytoplankton and other animals means more dead organisms • so more decomposition • releasing {methane / carbon dioxide} into the atmosphere • due to respiration by microorganisms • increasing global warming	(6)

		Additional guidance
Level 0	No awardable content	
Level 1	Demonstrates isolated elements of biological knowledge and understanding to the given context with generalised comments made. Vague statements related to consequences are made with limited linkage to a range of scientific ideas, processes, techniques and procedures. The discussion will contain basic information with some attempt made to link knowledge and understanding to the given context.	1 mark = one simple comment 2 marks = simple comments
Level 2	Demonstrates adequate knowledge and understanding by selecting and applying some relevant biological facts / concepts. Consequences are discussed which are occasionally supported through linkage to a range of scientific ideas, processes, techniques and procedures. The discussion shows some linkages and lines of scientific reasoning with some structure.	3 marks = simple discussion of one aspect 4 marks = detailed discussion of one aspect AND simple discussion of one other aspect OR simple discussion of three aspects
Level 3	Demonstrates comprehensive knowledge and understanding by selecting and applying relevant biological facts / concepts. Consequences are discussed which supported throughout by sustained linkage to a range of scientific ideas, processes, techniques and procedures. The discussion shows a well-developed and sustained line of scientific reasoning which is clear and logically structured.	5 marks = detailed discussion of one aspect AND simple discussion of two other aspects OR detailed discussion of two aspects 6 marks = detailed discussion of two aspects AND simple discussion of two other aspects OR detailed discussion of three aspects